`ww3recon`: reconstructing $E(f,\theta)$ from WW3 partition parameters

Technical note

2026 wave coupling project

2026-09-16, package version 0.2

1 Problem

WAVEWATCH III archives such as IOWAGA `GLOBMULTI_ERA5_GLOBCUR_01` store, at every grid cell and time, up to six spectral partitions $k = 0, \ldots, 5$ (0 = wind sea, 1–5 = swell systems ordered by energy) with five bulk parameters each, plus whole-spectrum integrals:

archive field	symbol	meaning	used for
$\texttt{phs}_k$	$H_s^{(k)}$	partition significant height [m]	partition energy $m_0^{(k)}$
$\texttt{ptp}_k$	$T_p^{(k)}$	partition peak period [s]	peak frequency $f_p^{(k)} = 1/T_p^{(k)}$
$\texttt{pdir}_k$	$\bar{\theta}^{(k)}$	partition mean direction (*coming-from*)	directional centre
$\texttt{pspr}_k$	$\sigma_\theta^{(k)}$	partition directional spread [deg]	directional width
$\texttt{pws}_k$	$w^{(k)}$	wind-sea fraction in the partition	family blend
`hs`, `t0m1`, `t02`	H_s, $T_{m0,-1}$, T_{m02}	whole-spectrum integrals	closure check, shape fit, tail fit

Storage is about 1/30 of the full spectrum. The task is to rebuild a directional variance density $E(f,\theta)$ on the native grid (36 frequencies, $f_i = 0.0339\,\text{Hz} \times 1.1^i$; 24 directions at 15°) that reproduces the archived integrals and is as close as possible to the archived 2D spectrum.

2 Grid, moments and conventions

Frequency bin widths follow WW3's own rule for a geometric ladder with ratio r,

$$\Delta f_i = \tfrac{1}{2} f_i \,(r - 1/r), \qquad \Delta\theta = 2\pi/24. \tag{1}$$

Spectral moments and the bulk parameters are discrete sums on this grid,

$$m_n = \sum_{i,j} f_i^n \, E(f_i,\theta_j)\, \Delta f_i \, \Delta\theta, \qquad H_s = 4\sqrt{m_0}, \quad T_{m0,-1} = \frac{m_{-1}}{m_0}, \quad T_{m02} = \sqrt{\frac{m_0}{m_2}}. \tag{2}$$

All archived integrals are truncated at the grid edge (0.95 Hz), so every moment in the fit is evaluated on the same truncated grid; comparing a reconstruction integrated on a finer or extended grid to an archived value is not a fair comparison.

Direction conventions. The archive's $\texttt{pdir}_k$, `dir` and `wnddir` are *coming-from* directions. The 2D spectrum axis is *going-to* (`sea_surface_wave_to_direction`, nautical degrees clockwise from north, stored in descending order). Partition directions are rotated $\bar{\theta}^{(k)} \mapsto \bar{\theta}^{(k)} + 180°$ before use, and the spectrum's direction axis is sorted ascending. Both are confirmed from file metadata, not inferred.

3 Model

Each partition is a separable product of a frequency shape and a directional distribution, and the partitions are summed:

$$E(f,\theta) = \sum_k m_0^{(k)}\, S^{(k)}(f)\, D^{(k)}(\theta), \qquad m_0^{(k)} = \frac{\left(H_s^{(k)}\right)^2}{16}, \tag{3}$$

with $S^{(k)}$ normalised so that $\sum_i S^{(k)}(f_i)\Delta f_i = 1$ and $D^{(k)}$ so that $\sum_j D^{(k)}(\theta_j)\Delta\theta = 1$. Consequently $m_0 = \sum_k m_0^{(k)}$ exactly, i.e. H_s is conserved partition by partition by construction, and the sum is invariant to the partition labelling, so index swapping between neighbouring cells does no harm. Absent partitions (NaN in the archive) are skipped.

3.1 Frequency shape

The shape of partition k is a blend of a wind-sea family S_{ws} and a swell family S_{sw}, both peaked at $f_p^{(k)}$ and normalised to unit m_0, weighted by the wind-sea fraction:

$$S^{(k)} = \omega^{(k)} S_{\mathrm{ws}}(f;\, f_p^{(k)},\, \Omega_c) + \left(1-\omega^{(k)}\right) S_{\mathrm{sw}}(f;\, f_p^{(k)},\, \lambda), \qquad \omega^{(k)} = \mathrm{clip}\Big(\frac{w^{(k)} - 0.3}{0.7 - 0.3},\, 0,\, 1\Big). \tag{4}$$

Blending the normalised shapes rather than switching on a threshold keeps time series continuous when `pws` crosses the threshold. Two shape parameters, Ω_c and λ, are shared by all partitions of a point and are the only free quantities; they are determined in Section 4.

Wind sea: Elfouhaily et al. (1997) long-wave branch (default). The gravity-wave part of the unified spectrum is a curvature spectrum $B_l(k)$ parameterised by the inverse wave age $\Omega_c = U_{10}/c_p$:

$$B_l(k) = \tfrac{1}{2}\,\alpha_p\,\frac{c_p}{c(k)}\,F_p(k), \qquad F_p = L_{PM}\,J_p\,\exp\Big[-\frac{\Omega_c}{\sqrt{10}}\Big(\sqrt{k/k_p}-1\Big)\Big], \tag{5}$$

$$L_{PM} = \exp\big[-\tfrac{5}{4}(k_p/k)^2\big], \qquad J_p = \gamma^{\Gamma}, \quad \Gamma = \exp\Big[-\frac{(\sqrt{k/k_p}-1)^2}{2\sigma^2}\Big], \quad \sigma = 0.08\,(1+4/\Omega_c^3), \tag{6}$$

$$\gamma = \begin{cases} 1.7 & \Omega_c \le 1 \\ 1.7 + 6\log_{10}\Omega_c & \Omega_c > 1 \end{cases}, \qquad \alpha_p = 0.006\sqrt{\Omega_c}, \qquad c(k) = \sqrt{\tfrac{g}{k}\left(1+(k/k_m)^2\right)},\ k_m = 370\,\mathrm{rad\,m^{-1}}. \tag{7}$$

It is mapped to frequency with the deep-water dispersion relation $k = (2\pi f)^2/g$ and Jacobian $\mathrm{d}k/\mathrm{d}f = 8\pi^2 f/g$: $S_{\mathrm{ws}}(f) \propto B_l(k)\,k^{-3}\,\mathrm{d}k/\mathrm{d}f$, with k_p set from $f_p^{(k)}$. Because the shape is renormalised to $m_0^{(k)}$, the Elfouhaily amplitude prediction is discarded and only the shape survives: a JONSWAP-like form whose peak enhancement and width are tied to one physical parameter, $\Omega_c \in [0.84, 5]$, with the f^{-4}-to-f^{-5} transition of the high-frequency slope built in. The capillary branch B_h peaks at ∼96 Hz and contributes nothing on the native grid. Alternatives available in the package: JONSWAP (γ), Donelan (γ, f^{-4}), Pierson–Moskowitz (no parameter); these three carry a free tail exponent n (Section 4.1).

Swell: single Ochi–Hubble component (default). With $x = f/f_p^{(k)}$ and shape parameter $\lambda \in [1, 12]$,

$$S_{\mathrm{sw}}(f) \propto x^{-(4\lambda+1)}\exp\Big[-\frac{4\lambda+1}{4}\,x^{-4}\Big], \tag{8}$$

which reduces to Bretschneider at $\lambda = 1$, sharpens as λ grows, keeps a power-law high-frequency tail and is strictly positive. A Gaussian in f is also available; on real WW3 spectra it fails (spectral correlation ≈ 0.69 versus 0.90), because the sum of narrow Gaussians is a comb with gaps between partitions whereas WW3 spectra are smooth. Ochi–Hubble's tails fill those gaps.

3.2 Directional distribution

Each partition uses a cosine-power distribution centred on its mean direction:

$$D^{(k)}(\theta_j) = \frac{\cos^{2s_k}\left(\frac{1}{2}\,\Delta_j\right)}{\sum_{j'} \cos^{2s_k}\left(\frac{1}{2}\,\Delta_{j'}\right)\Delta\theta}, \qquad \Delta_j = \mathrm{wrap}_{(-\pi,\pi]}\left(\theta_j - \bar{\theta}^{(k)}\right), \tag{9}$$

normalised numerically on the 24 bins (the analytic Γ-function constant leaves percent-level energy errors for narrow swell on so coarse a grid). The exponent follows from the archived spread by inverting the first-moment (Kuik) relation for this family: with $\langle\cos\Delta\rangle = s/(s+1)$ and $\sigma_\theta^2 = 2(1 - \langle\cos\Delta\rangle)$,

$$s_k = \frac{2}{\left(\sigma_\theta^{(k)}\right)^2} - 1, \qquad s_k \in [0.5, 200]. \tag{10}$$

Thus each partition's first directional moment (a_1, b_1) matches its archived direction and spread in the continuum limit; on the discrete grid the match is approximate.

The wrap in (9) matters. Without it, a partition centred at 330° sees the 0° bin as an offset of $-330°$, $\cos(-165°) < 0$ is clipped to zero and every bin across the 0/360 seam receives no energy from that partition. Renormalisation hides the loss from H_s but not from the spectral shape. This was present in the first real-data run and fixed on 2026-09-16; it moved the extratropical site from 0.869 to 0.896 spectral correlation.

4 Fitting the shape parameters

The archive provides one more integral than (3) uses: $T_{m0,-1}$. Since m_{-1} is linear in the partition energies and, for a unit-m_0 shape, the ratio

$$\rho_{\mathrm{ws}}(f_p, \Omega_c) = \frac{m_{-1}}{m_0}\Big|_{S_{\mathrm{ws}}}, \qquad \rho_{\mathrm{sw}}(f_p, \lambda) = \frac{m_{-1}}{m_0}\Big|_{S_{\mathrm{sw}}} \tag{11}$$

depends only on the peak frequency and the shape parameter, the constraint reads

$$\underbrace{\sum_k m_0^{(k)}\,\omega^{(k)}\,\rho_{\mathrm{ws}}\left(f_p^{(k)}, \Omega_c\right)}_{A_{\mathrm{ws}}(\Omega_c)} + \underbrace{\sum_k m_0^{(k)}\,\left(1-\omega^{(k)}\right)\rho_{\mathrm{sw}}\left(f_p^{(k)}, \lambda\right)}_{A_{\mathrm{sw}}(\lambda)} = T_{m0,-1}\sum_k m_0^{(k)} \equiv \mathcal{T}. \tag{12}$$

The right-hand side uses the *sum of the partition energies*, not the archived m_0, so that the shape fit is decoupled from any energy the partitioning failed to capture (closure $\sum_k (H_s^{(k)})^2/H_s^2$ is reported separately as a diagnostic; it is 1.000 for IOWAGA). ρ_{ws} and ρ_{sw} are tabulated once on a (f_p, parameter) grid (96 log-spaced f_p values, 45 parameter values) on the truncated native grid, so that truncation is included; the per-point fit is then a table lookup and a vectorised search rather than an optimiser call.

One equation for two unknowns is under-determined, so weak quadratic priors break the degeneracy:

$$(\Omega_c, \lambda) = \arg\min\ \left(\frac{A_{\mathrm{ws}}(\Omega_c) + A_{\mathrm{sw}}(\lambda) - \mathcal{T}}{\epsilon\,|\mathcal{T}|}\right)^2 + \Lambda\left(\frac{\Omega_c - \Omega_c^0}{\Delta\Omega_c}\right)^2 + \Lambda\left(\frac{\lambda - \lambda^0}{\Delta\lambda}\right)^2, \tag{13}$$

with $\epsilon = 0.01$ (relative tolerance on $T_{m0,-1}$), $\Lambda = 0.7$, priors $\Omega_c^0 = 1.5$, $\lambda^0 = 3$, and $\Delta(\cdot)$ the parameter range, which makes Λ dimensionless and family-independent. The minimum is found by exhaustive search on the tabulated grid. Where a point contains only one family the fit is exactly determined and the other prior is inert; such points are flagged `underconstrained_ws/sw`. Because there are exactly as many constraints as free parameters, the priors are load-bearing, and matching $T_{m0,-1}$ does not by itself prevent an unphysical shape—this is why the choice of swell family, not the fit, dominates the skill.

4.1 Tail exponent

For wind-sea families with a free high-frequency exponent n ($S \propto f^{-n}$; JONSWAP, Donelan, Pierson–Moskowitz), a second stage picks $n \in \{3.5, 4, \ldots, 6\}$ to match the archived T_{m02} through $m_2 = m_0/T_{m02}^2$, which is tail-weighted and therefore carries the information the first stage does not use. Stages alternate twice. For the Elfouhaily default the slope is intrinsic and this stage is skipped. With either choice the reconstructed $E(f)$ runs ∼50% low above 0.3 Hz at all validated sites; this is the largest open error source for tail-weighted quantities such as Stokes drift.

5 Assembly and conserved quantities

With (Ω_c, λ) fixed, the spectrum follows from (3). What is conserved:

- m_0 per partition and in total (exact, by normalisation), hence H_s up to the archive's own partition closure;
- $T_{m0,-1}$ to the tolerance ϵ where the priors do not dominate;
- T_{m02} approximately, when a free tail exponent is available;
- per-partition mean direction and first-moment spread (continuum-exact, discretely approximate).

Not constrained: the whole-spectrum `dir`/`spr`, the second directional moments (a_2, b_2), and the shape of the high-frequency tail above ∼0.3 Hz.

6 Validation summary

Self-partition mode (partitions derived from the truth spectrum by a local watershed, so the numbers are the method's intrinsic ceiling without grid-to-station sampling error), January 1993, 248 three-hourly steps per site, Elfouhaily + Ochi–Hubble:

site	spectral corr. (mean)	corr. p10	H_s scatter index
Tropical Eq. Pacific (0N 140W)	0.917	0.877	1.1%
Extratropical N Atlantic (52N 30W)	0.896	0.830	0.15%
Southern Ocean deep (58S 90W)	0.890	0.838	0.31%
N California offshore (38N 124W)	0.901	0.830	0.45%

The spectral correlation is $\langle E_{\rm rec}, E_{\rm true}\rangle / \|E_{\rm rec}\| \|E_{\rm true}\|$ with the $\Delta f \Delta\theta$ inner product. The four wind-sea families lie within 0.004 of each other; replacing Ochi–Hubble by a Gaussian swell costs about 0.2. With the archive's own partitions read from the gridded monthly file (`example_reconstruct_from_nc.py`) the same month gives 0.885 (p10 0.821) at the extratropical site and 0.910 (p10 0.863) at the tropical site; the gap to the self-partition numbers is the grid-cell-to-station sampling penalty plus the difference between WW3's partitioning and the local watershed. H_s and $T_{m0,-1}$ are reproduced to $< 0.1\%$ in both modes.

7 Known limitations

- Deep water only: $k = (2\pi f)^2/g$ throughout.
- The a_1-based mapping from `pspr` to s has not been confirmed against the archive's definition; if `pspr` is a_2-based the exponent should be obtained by numerical inversion.
- Tail deficit above 0.3 Hz (see Section 4.1).
- `ptp` is quantised to the native frequency bins, which limits narrow swell.

References

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